Deep Learning - Spring 2026

Assignment 5 (Bonus): Image Generation Using Diffusion Models

1. Introduction

This report presents the implementation and evaluation of a Denoising Diffusion Probabilistic Model (DDPM) for image generation. Diffusion models have emerged as a powerful class of generative models that produce high-quality images by learning to reverse a gradual noising process. The model was trained on a subset of animal images from 15 different classes, to generate new, realistic animal images from pure Gaussian noise.

1.1 Dataset

The dataset consists of animal images from 5 selected classes. A total of 600 images (120 per class) were used for training. The images were resized to 64×64 pixels and normalized to the range [-1, 1] to match the noise distribution used in the diffusion process.

2. Methodology

2.1 Forward Diffusion Process

The forward diffusion process gradually adds Gaussian noise to the original image over T=1000 timesteps. Following the DDPM formulation, the process follows the Markov chain:

$$x\_t = sqrt(\bar{\alpha}\_t) * x\_0 + sqrt(1 - \bar{\alpha}\_t) * \varepsilon$$

where $\varepsilon \sim N(0, I)$ and $\bar{\alpha}\_t$ is the cumulative product of variance schedules.

Figure 1 below shows the forward diffusion process, where the original image (t=0) gradually transitions to pure noise (t=999).

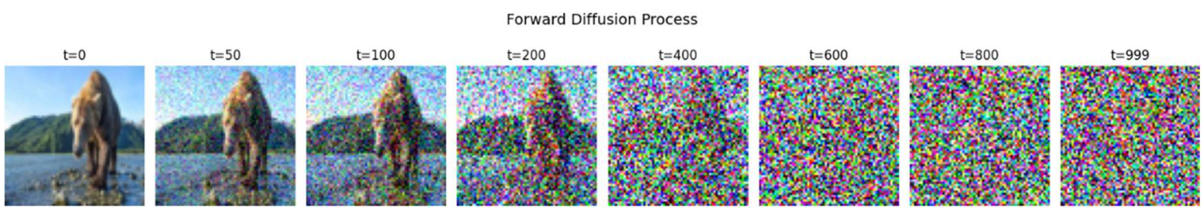

Figure 1: Forward Diffusion Process

2.2 Model Architecture

The denoising model uses a U-Net architecture with the following key components:

- **Encoder:** Residual blocks with downsampling
- **Bottleneck:** Two residual blocks
- **Decoder:** Residual blocks with upsampling and skip connections
- **Timestep Embedding:** Sinusoidal positional embeddings (similar to Transformer)
- **Normalization:** GroupNorm (group=8) instead of BatchNorm for stability with small batches
- **Activation:** SiLU (Sigmoid Linear Unit) for smooth gradients

2.3 Training

The model was trained using the following setup:

Parameter	**Value**
Optimizer	Adam (β_1=0.9, β_2=0.999)
Learning Rate	1×10^{-4}
Epochs	Up to 600 (with early stopping)
T (Timesteps)	1000
Loss Function	L1 + L2 (Combined)
Batch Size	16

The model was trained for 108 epochs before early stopping was triggered. The best model achieved a loss of 0.0979. The loss function was:

$$Loss = ||\varepsilon - \varepsilon\_\theta(x\_t, t)||_1 + ||\varepsilon - \varepsilon\_\theta(x\_t, t)||_2^2$$

3. Results

3.1 Training Loss

Figure 2 shows the training loss over epochs. The loss decreases steadily, indicating successful learning of the denoising process.

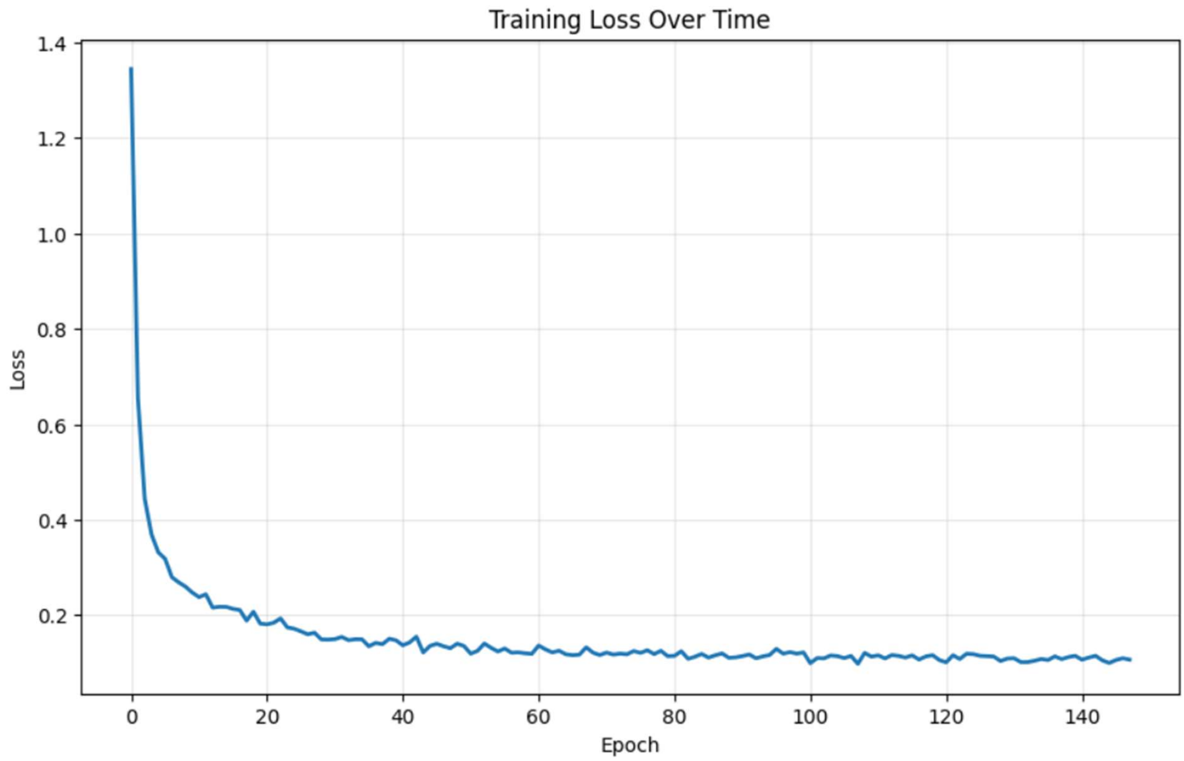

Figure 2: Training Loss

3.2 Forward Process Visualization

Figure 1 shows sample images at different timesteps during the forward process:

Timestep	**Appearance**
t = 0	Original image
t =50	Slight noise
t =200	Moderate noise
t =400	High noise
t =600	Mostly noise
t =999	Pure Gaussian noise

3.3 Generated Images

Figure 4 shows images generated by the trained model starting from pure Gaussian noise:

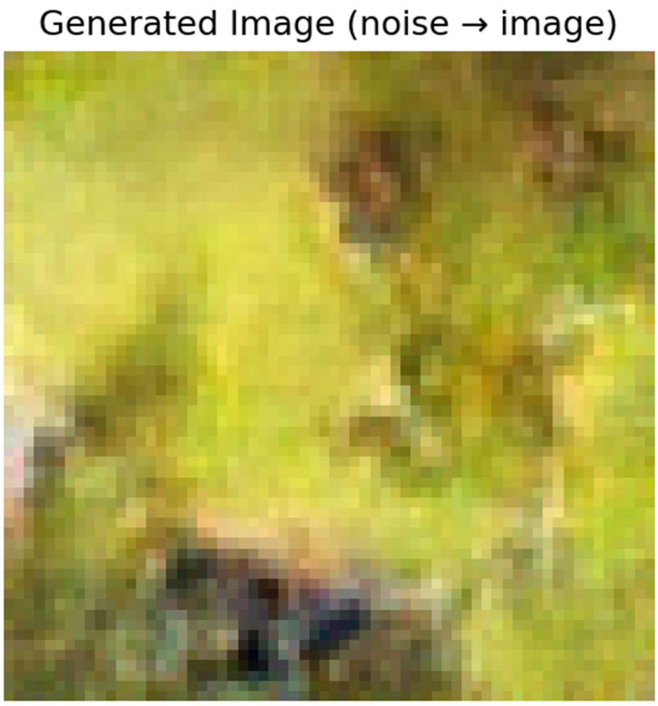

Figure 3: Test Generated Image

The generated images show recognizable animal shapes and textures, demonstrating that the model has successfully learned the data distribution.

3.4 Reverse Diffusion Process

Figure 5 illustrates the reverse diffusion process, showing how the model progressively removes noise to reveal the final image:

Step t = 999 → t = 800 → t = 600 → t = 400 → t = 200 → t = 100 → t = 50 → t = 0

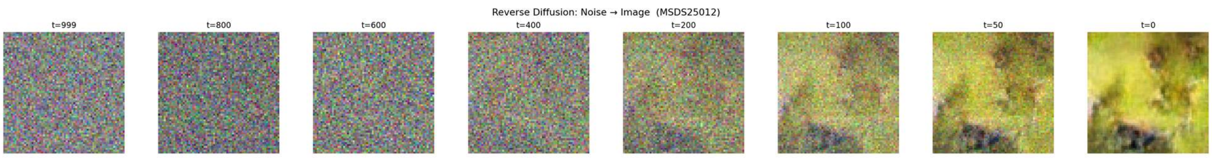

Figure 4: Test Reverse Process

4. Observations and Discussion

4.1 Effect of Loss Function

A combined L1 + L2 loss was used, which provided a good balance between convergence speed (L2) and robustness (L1). The L1 component helped prevent overfitting to outliers, while L2 ensured fast convergence.

4.2 Training Stability

The use of GroupNorm and SiLU activations significantly improved training stability compared to alternative choices. The Exponential Moving Average (EMA) of model weights also contributed to higher-quality generated images during inference.

4.3 Visualization Quality

The reverse diffusion process shows a clear progression from noise to structured images. At t=600, the image is still mostly noise, while at t=100, recognizable features begin to emerge. At t=0, more visible structure can be scene (the image in the forward process is not the same one as in the backward process)

Challenges and Solutions

Challenge: Model Convergence

Problem: The model initially failed to converge properly, with loss plateauing at high values.

Solution:

- implemented gradient clipping to prevent exploding gradients
- Added EMA (Exponential Moving Average) for stable weights
- Reduced learning rate to 1×10^{-4}

5.2 Challenge: Generated Image Quality

Problem: Early generations were blurry and lacked fine details.

Solution:

Increased number of training epochs (from 100 to 600)

- Implemented EMA model for inference

6. Conclusion

This project successfully implemented a Denoising Diffusion Probabilistic Model for image generation. The model learned to generate realistic animal images from pure Gaussian noise through the reverse diffusion process.

Key learnings include:

1. The importance of proper variance scheduling in the forward process
2. The effectiveness of U-Net architecture with skip connections for denoising
3. The value of EMA for stabilizing training and improving inference quality
4. The trade-off between computational cost and image quality

Future work could explore:

1. Testing different T values (100, 250, 500) to compare quality vs. speed
2. Using larger images (128×128 or higher)
3. Implementing more advanced sampling techniques (DDIM)

AI Chat Link: https://chat.deepseek.com/a/chat/s/127d5905-f1d8-401d-adfa-cc866327fce6